

Pinned group summary: $SU(n=4, q=2, \text{eps}=-1)$

Pinning-Sympy automated output

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1 Basic attributes

Group	SU(n=4, q=2, eps=-1)
Matrix size	4×4
Rank	2

1.1 Skew-hermitian form

Dimension	4
Witt index	2
Primitive element	$\sqrt{d}$
Epsilon	-1

Matrix:
$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

1.2 Generic Lie algebra element

$$\begin{bmatrix} \sqrt{d}x_{i00} + x_{r00} & \sqrt{d}x_{i01} + x_{r01} & -x_{r02} & -\sqrt{d}x_{i12} + x_{r12} \\ \sqrt{d}x_{i10} + x_{r10} & -\sqrt{d}x_{i00} + x_{r11} & \sqrt{d}x_{i12} + x_{r12} & -x_{r13} \\ -x_{r20} & -\sqrt{d}x_{i30} + x_{r30} & \sqrt{d}x_{i00} - x_{r00} & \sqrt{d}x_{i10} - x_{r10} \\ \sqrt{d}x_{i30} + x_{r30} & -x_{r31} & \sqrt{d}x_{i01} - x_{r01} & -\sqrt{d}x_{i00} - x_{r11} \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} \end{bmatrix}$$

1.4 Trivial characters

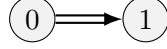
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	B2
Reduced	True
Simply laced	False
Number of roots	8

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 & -2 \\ -1 & 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
$(-2, 0, 0, 0)$	$(-1, 0, 1, 0)$	No	—	No
$(-1, -1, 0, 0)$	$(-1, -1, 1, 1)$	No	—	No
$(-1, 1, 0, 0)$	$(-1, 1, 1, -1)$	No	—	No
$(0, -2, 0, 0)$	$(0, -1, 0, 1)$	No	—	No
$(0, 2, 0, 0)$	$(0, 1, 0, -1)$	Yes	+	No
$(1, -1, 0, 0)$	$(1, -1, -1, 1)$	Yes	+	No
$(1, 1, 0, 0)$	$(1, 1, -1, -1)$	No	+	No
$(2, 0, 0, 0)$	$(1, 0, -1, 0)$	No	+	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

α	β	Pairs (i, j)
$(-2, 0, 0, 0)$	$(1, -1, 0, 0)$	$(1, 1), (1, 2)$
$(-2, 0, 0, 0)$	$(1, 1, 0, 0)$	$(1, 1), (1, 2)$
$(-1, -1, 0, 0)$	$(-1, 1, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0)$	$(0, 2, 0, 0)$	$(1, 1), (2, 1)$
$(-1, -1, 0, 0)$	$(1, -1, 0, 0)$	$(1, 1)$
$(-1, -1, 0, 0)$	$(2, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 1, 0, 0)$	$(0, -2, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 1, 0, 0)$	$(1, 1, 0, 0)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(2, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, -2, 0, 0)$	$(1, 1, 0, 0)$	$(1, 1), (1, 2)$
$(0, 2, 0, 0)$	$(1, -1, 0, 0)$	$(1, 1), (1, 2)$
$(1, -1, 0, 0)$	$(1, 1, 0, 0)$	$(1, 1)$